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Early Disease Detection in Plants using CNN

Tejaswini[a,*], Priyanka Rastogi[a], Swayam Dua[a], Manikanta[a], Vikas Dagar[a]

[a] *Manav Rachna International Institute of Research & Studies, Faridabad, Haryana, India*

Abstract

The Indian economy benefits from early disease detection in plant leaves. According to reports, 10-30% of crops suffer harm from diseases that are not discovered during the curing process. Recent advancements in deep learning, particularly Convolutional Neural Networks (CNNs), have paved the way for transformative solutions in disease identification for agriculture. This study addresses the critical issue of early disease detection by harnessing the power of deep learning models, specifically CNNs. Different leaf disease detection technologies are used for different crops. The pre-trained deep learning model is used in this study to identify and categorize leaf diseases. A dataset of tomato, potato, and bell pepper leaf pictures from the plant village repository was employed for the current investigation. The developed model can detect 12 plant diseases in normal leaf tissue. The quantitative assessment of our CNN-based technique reveals an impressive accuracy rate of 86.21%. This notable accuracy underscores the efficacy of our approach in the challenging domain of plant disease detection. Our findings show potential for the larger agricultural environment beyond the immediate quantitative advantages. This technical advance not only paves the way for improved crop output and lower losses, but also brings in a new era of data-driven sustainability in agriculture.The bottom line is that our research "offers a tangible pathway for leveraging AI-powered solutions to address the long-standing challenges of plant disease detection, thereby significantly contributing to the well-being of farmers and the sustenance of the Indian agricultural sector.

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Keywords: Deep Learning; classification; CNN; leaf disease detection.

1. Introduction

India is a developing nation where agriculture supports around 70% of the people. Plant disease results in a significant decline in the quantity and quality of agricultural goods. Therefore, to ensure food security, both early disease detection and the development of better crop varieties is essential.

* Corresponding author. Tel.: + 91 7986799578
E-mail address: tejaswinigupta606@gmail.com

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The traditional technique of disease identification has been manual examination by farmers or professionals, but this may be time-consuming and expensive, making it impracticable for millions of small and medium-sized farms worldwide. The development of automated systems for plant disease identification by checking for telltale signals on leaves has been sped up by the construction of computer vision models. These methods attempt to make the detecting process as straightforward and error-free for farmers as possible.

Studies have concentrated on employing image processing and feature extraction to build methods for disease detection. In order to recognise and categorise plant diseases, numerous machine learning models have been used. Since deep learning (DL) models have been developed, this area of research currently seems to offer a lot of potential for increased accuracy. Numerous customised DL architectures are used with various visualisation approaches to detect and classify the symptoms of plant diseases. In addition, various performance metrics are used to evaluate these frameworks and procedures. Teaching a neural network to understand patterns and symptoms shared by various ailments is one step in the deep learning process for detecting plant diseases. The deep learning model gains the ability to distinguish between healthy and unhealthy plants by examining plant photos, making automated detection possible. By doing away with manual inspection, this method yields quicker and more precise results. Convolutional neural networks and other deep learning algorithms extract pertinent elements from the photos, allowing the model to categorise the state of the plant's health. By enabling early identification and intervention, this technology has the potential to revolutionise agricultural practices, resulting in higher crop yields and decreased economic losses.

Thus, this research paper intends to build a plant disease classifier for three plant species. These include the tomato (*Solanum Lycopersicon*), the pepper bell (*Capsicum annuum*), and the potato (*Solanum tuberosum*). The model is trained to recognize a small set of diseases or health conditions specific to each species. The model predicts the disease in these plants by using the proposed model. The treatment to stop the spread of the disease in other parts of the plant can be stopped by the farmer, and further corrective measures can be taken by him in a timely manner. The research objectives of this paper are: i) The model's general suitability for categorizing diseases using leaf images. ii) Evaluate the model performance iii) Use the trained model to deploy a user-friendly mobile application.

The rest of the paper is structured into five sections: the literature survey, proposed methodology and experimental setup, results and discussion, conclusion and future scope. Section II reviews existing research on plant disease detection techniques and identifies the need for more accurate and efficient methods. Section III outlines the use of CNN for disease detection in plants, explaining the architecture and training process. Section IV The experimental setup describes the dataset used, data preprocessing techniques, and evaluation metrics employed. Section V The results and discussion section presents the performance of the CNN model in terms of accuracy, precision, and recall, along with a detailed analysis of the findings. Section VI summarizes the key contributions of the study, highlighting the effectiveness of CNN for plant disease detection and its potential impact on agriculture. Finally, the future scope suggests further research directions, including exploring transfer learning, real-time disease detection, and integrating other advanced machine learning techniques. This research paper aims to advance the field of plant disease detection and provide valuable insights for future studies in this area.

2. Literature Survey

A preliminary literature survey was conducted for plant disease detection in which existing research and studies related to the detection and diagnosis of plant diseases were reviewed. The survey aimed to gather information on the methods, techniques, and algorithms used by researchers in this field. Researchers have used various techniques, including spectral data for hyperspectral imaging, machine learning algorithms like support vector machines (SVM) and deep learning models like CNNs, and image processing techniques like segmentation and feature extraction. These techniques were used to precisely diagnose and categorize illnesses by analysing spectral data or plant image analysis.

Saleem et al. (2019) [2] proposed plant disease detection and classification using deep learning methods. It was a pioneering effort that used hyperspectral imaging to identify plant disease automatically. The experimental study aimed to detect Tomato Spotted Wilt Virus (TSWV) infection in capsicum plants. After inoculation, a routine automated system was developed to collect hyperspectral hypercubes of plant leaves in the VNIR and SWIR spectrum

regions. Additionally, the technology was used to collect pertinent meta-data about the plants. After processing the data, it was possible to see how effectively plant leaves could be automatically differentiated using machine learning and image processing techniques. The results showed an accuracy of over 90%. They took advantage of the VNIR (400–1000 nm) and SWIR (900–2500 nm) portions of the electromagnetic spectrum.

Durmuş et al. (2017) [3] proposed disease detection on the leaves of tomato plants by using deep learning. The dataset of tomato leaf images was downloaded from the Plant Village Dataset, and the proposed research was based on the six different types of tomato leaf diseases. The error-correcting output code was utilised to train a deep convolution neural network model, which was employed for detection and classification. The classification results were computed using parameters including accuracy, specificity, sensitivity, F-Score, and true negative rate. The trained model was compared to the basis paper and had a 98.8% accuracy rate.

Sharma et al. (2020) [4] proposed a performance analysis of deep-learning CNN models for plant disease detection using image segmentation. This study has demonstrated the viability of training a CNN model instead of whole images using segmented and annotated images. Model performance on independent data increases from 42.3% to 98.6% when the same CNN model was trained using segmented pictures (S-CNN) instead of full images (F-CNN). With 82% of the test dataset displaying an increase in confidence, quantitative examination of self-classification confidence demonstrated a considerable improvement. Before CNN model training, picture preprocessing has proven invaluable in achieving excellent real-world performance.

Khirade et al. (2015) [5] proposed an approach to Plant disease detection using image processing. Various novel approaches to quickly identify and categorise any illness are essential for the successful production of crops and for ensuring their sustainability. Image Processing is one of the modern technologies that paved the path for this. This study investigated the segmentation of the sick plant components utilising several techniques. This study also examined the methods for feature extraction and classification and the classification of plant diseases. Plant diseases can be effectively classified using SVMs, backpropagation methods, self-organizing feature maps, and other ANN techniques. These techniques enable accurate diagnosis and classification of numerous plant diseases using image processing tools.

Sood et al. (2020) [6] describe a hybrid system for detecting and classifying plant disease using qualitative texture features analysis. In this work, the diseased affected capsicum piece is first subdivided, and then the features of the infected region are extracted using feature extraction. The classification of the capsicum illnesses is completed using an SVM classifier. In order to distinguish between healthy and diseased capsicum and its leaves, the proposed solution is tested for the five illnesses anthracnose, bacterial spot, powdery mildew, Cercospora leaf spot, and grey leaf spot. An SVM classifier can differentiate between healthy and sick capsicum with almost 100% accuracy. Using multi-class classifiers, diseased capsicum can be further categorised based on the type of disease, either bacterial or fungal. According to the suggested method, there are two types of healthy and diseased capsicums. This investigation is limited to identifying five capsicum infections at this time.

Devi et al. (2019) [7] proposed IoT-enabled efficient detection and classification of plant diseases for agricultural applications. The proposed IoT-enabled sickness detection and categorization system was implemented in the openCV-python environment. Eighty images of the hill banana plant were used to test and show the process. Two diseases—bunchy top of the banana and sigatoka leaf spot disease—are chosen for demonstration and evaluation. During the image acquisition stage, the plant images were shrunk to 256 x 256, and the histogram was levelled using a histogram equalisation technique. Once the segmentation process had been finished using the K-means clustering method, the GLCM features were uploaded to the cloud. The experts gathered the information, looked through it, and provided the farmers with solutions. RFC generated a classification accuracy of approximately 99% when compared to the current k-NN classifier in terms of system performance.

Singh et al. (2021) [8] presented disease detection and classification of potato plant leaves using machine learning methodologies. Utilising machine learning techniques, the proposed model was used to identify and categorise potato leaves that were damaged and those that weren't. The entire procedure involved several processes, including image collecting, image preprocessing, image segmentation, feature extraction, evaluation of the impacted region,

processing, training and testing data, and assessment through metrics. The Grey Level Co-occurrence Matrix was utilised for feature extraction, the K-means approach was employed for image segmentation, and a multi-class support vector machine algorithm with a linear kernel function was used to classify potato leaves. The proposed framework was assessed using specific assessment criteria, such as precision, recall, F1-score, and accuracy. The proposed model's F1-score is approximately 96.16%, its precision is roughly 96.12%, its recall is approximately 96.25%, and its total accuracy is approximately 95.99%.

3. Proposed Methodology

Figure 1 shows the entire process of the proposed model. The three main steps of the technique are the data preprocessing stage, model training using the training set, and prediction using the test set. The following subsections cover the specifics of each phase.

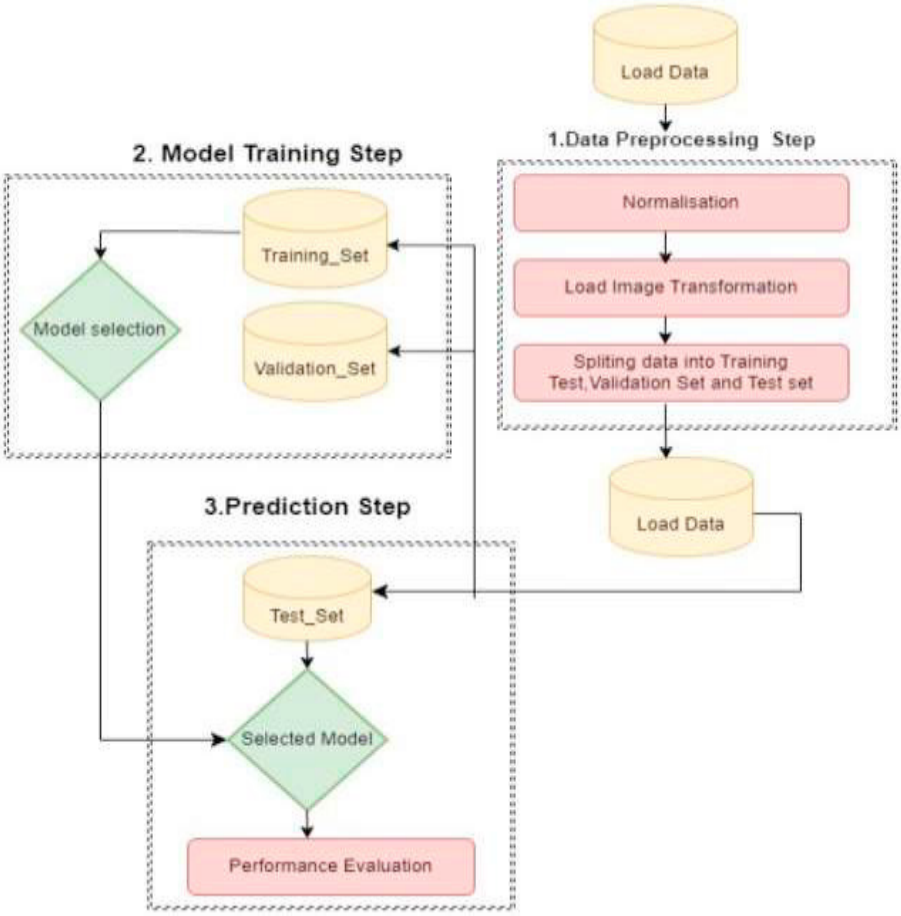


Fig. 1. Overall Proposed Methodology.

3.1. Data Preprocessing Step

The data preprocessing step involved the normalization of images. Image normalisation is a preprocessing procedure used in the dataset for deep-learning plant disease detection to guarantee that all images have uniform sizes and ranges of pixel values. This stage is crucial for boosting the model's convergence during training and for enhancing its generalisation abilities. Image normalisation frequently involves the following steps:

Rescaling: Images are scaled or resized to 224x224 pixels. Usually, this is done to ensure that all of the images are

the same size so that feeding them to the neural network is simpler.
Pixel Value Normalisation: The image's pixel values are altered to fall between a range of values, frequently between 0 and 1 or -1 and 1. This normalisation aids in minimising the effects of variations in pixel intensity and enhances the neural network's convergence during training.
Mean Subtraction: Each pixel has its mean pixel value from the entire dataset is subtracted. This process prevents neural network activation functions from becoming saturated and centres the data distribution around zero.
Standardisation: The pixel values are further scaled by dividing the pixel values by the dataset's standard deviation. The unit variance of the data is ensured using this procedure, which also speeds up neural network learning.

The dataset is divided into three sets: the training set, the validation set, and the test set once it has been normalised.

Training Set: Most of the dataset, known as the training set, is used to train the deep learning model. Through forward and backward passes throughout the training phase. It is used to update the model's parameters.
Validation Set: Using the validation set, the validation performance of the model is assessed, and its hyperparameters are modified. As it enables tracking of the model's performance on data it hasn't seen during training, it is crucial for preventing overfitting.
Test Set: The test set is a distinct dataset that is only utilized following the conclusion of model training. It measures the trained model's ultimate performance and ability to generalize to new data.

Data augmentation generates new images that are altered versions of our input data by applying image modification, contrast adjustments, and image blurring without affecting the labels on the original photos. The training set images were augmented for the model's robustness. The data augmentation steps are depicted in Figure 2.

Even the issue of over-adjustment is also decreased by the expansion of the entire “train”. To boost data, the “Keras Image data Generator” class was used, and the following options produced the best results:

(a) Rotation range: 20

(b) Width and height shift ranges: 0.2

(c) Horizontal flip: True

(d) Vertical mode: True

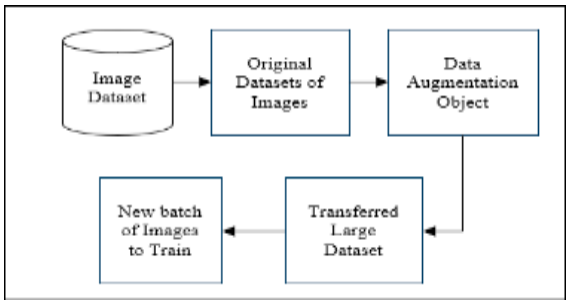


Fig. 2. Data Augmentation Steps.

3.2. Proposed Model Architecture

The architecture of CNN, which enables the effective processing of image data, is depicted in Figure 3. A deep CNN architecture is made up of numerous layers of various kinds. It is the most often used technique for sifting through large datasets and extracting useful information. The multiple layers of the proposed CNN model architecture for plant disease detection are intended to extract features from the input photos and classify them into several disease categories. The architecture follows a sequential structure, with each layer performing specific operations to transform the input data. The details of CNN model layers are described in brief.

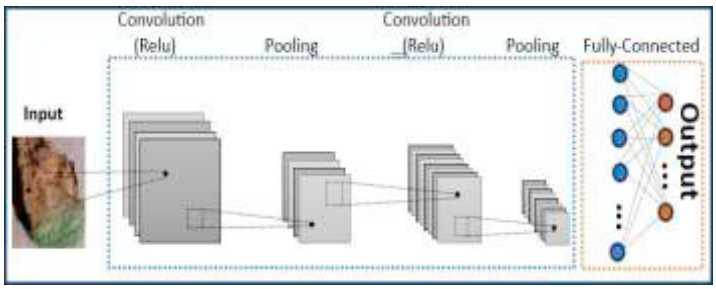


Fig. *3* Convolution Neural Network (CNN) Architecture

Convolutional layers

Convolutional layers make up the first few layers of the CNN. Convolutional layers perform feature extraction by applying filters to the input image. These filters learn spatial patterns and local features present in the image. From one block to another, the number of filters gradually increases, but the size of the filter remains fixed at 3 x 3. There are 32 filters in the first convolutional block, 64 in the second, and 128 in the third. The size of the feature maps was lowered due to the use of pooling layers in each block, necessitating an increase in the number of filters.

Convolutional layers are particularly effective in capturing hierarchical representations of images, allowing the model to learn discriminative features at different scales. Each convolutional layer applies a set of filters (kernels) to the input image. Each filter convolves with the image, producing feature maps highlighting specific patterns. Non-linear activation functions, such as ReLU (Rectified Linear Unit), are commonly applied after each convolutional layer to introduce non-linearity and increase the model's ability to capture complex patterns.

Pooling Layer

A max pooling layer with a pool size of 2x2 is added after each convolutional layer. The feature maps created by the convolutional layers are downsampled using pooling layers. Doing this makes it possible to keep key features while reducing the feature maps' spatial dimensions. By condensing local data, pooling layers simplify computation and increase the model's spatial invariance.

The most popular pooling process, max pooling, chooses the highest value possible within a predetermined pooling frame. Pooling layers increases the model's resistance to changes in object location, scale, and rotation by downsampling the feature maps.

Flattened Layer

The flattened layer creates a 1-dimensional vector from the output of the preceding levels. This flattening phase is essential before transferring the data into a fully connected layer.

Fully Connected Layers

The CNN architecture's fully connected layers, which are found near the bottom, are in charge of categorising the learned features into particular illness categories. Fully connected layers capture global relationships between features and enable the model to make predictions based on the extracted representations. One or more fully connected layers receive the output of the final pooling layer after it has been vectorized and flattened. Fully Connected layers provide thorough feature combining and categorization since every neuron in the layer below is coupled to every neuron in the layer above.

The first dense layer consists of 64 neurons with the ReLU activation function. It captures global relationships between features. The second dense layer has 16 neurons with the ReLU activation function, allowing further feature combinations. The 15 neurons in the last dense layer, which have a softmax activation function and provide a probability distribution over the 15 illness classifications, are what make up this layer. The softmax activation function ensures that the predicted probabilities sum up to 1.

4. Dataset and implementation details

4.1 Dataset Creation

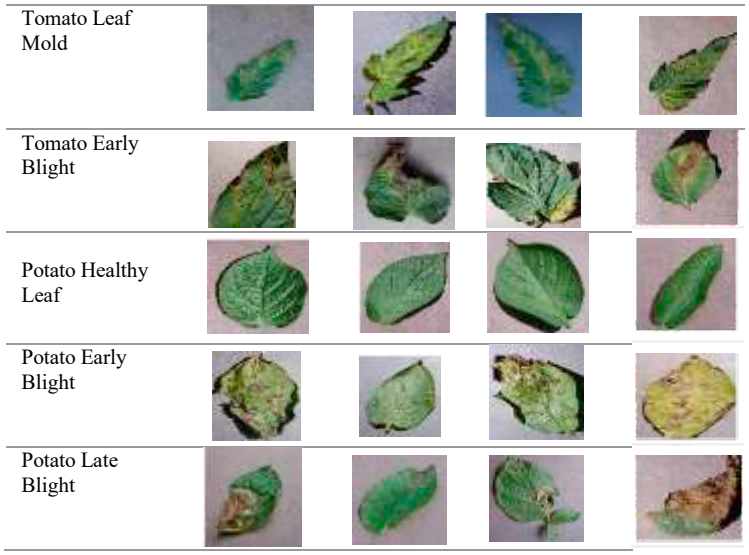
The 'Plant Village Dataset' is the source of all Potato, Tomato, and Pepper Bell pictures. The 20,639 leaf images in the Plant Village dataset are categorised into 15 groups based on species and illness, as shown in Table 1. The leaf pictures dataset includes both healthy and ill leaves that have been impacted by various biotic causes. Images of every significant type of leaf disease are included in this dataset. These vegetables are part of the staple diet in many Indian households. Thus, plant disease detection in such plants was important. We build the model using a subset of 70% and test it using a subset of 30%. Examples of the "Train" folder images from the adopted distribution are included in Table 2.

Table 1. Dataset Used for Classification

Species	Class	No. of Images
Pepper Bell	Healthy	1478
Pepper Bell	Bacterial Spot	997
Potato	Early Blight	1000
Potato	Late Blight	1000
Potato	Healthy	152
Tomato	Bacterial Spot	2172
Tomato	Early Blight	1000
Tomato	Late Blight	1909
Tomato	Leaf Mold	952
Tomato	Septoria leaf Spot	1771
Tomato	Spider mites	1676
Tomato	Target Spot	1404
Tomato	Yellow Leaf Curl Virus	3209
Tomato	Healthy	1591
Tomato	Mosaic Virus	373

Table 2. Images from Dataset

Pepper Bell Healthy Leaf				
Pepper Bell Bacterial Spot				
Tomato Healthy Leaf				



4.2 Implementation Details

The Adam optimizer, a stochastic gradient descent (SGD) variant that modifies the learning rate during the course of training, is used to train the model. The Adam optimizer's default setting of a negligibly low number, such as 0.001, is utilized as the learning rate. A loss function suitable for multi-class classification applications is used, categorical cross-entropy. The training process lasts for 20 epochs, and to improve the model's capacity for generalization, data augmentation is applied during training using the ImageDataGenerator object. Accuracy, which measures the percentage of accurate predictions overall predictions, and categorical cross-entropy loss, which gauges the discrepancy between the actual and predicted probability distributions, are the performance metrics considered for evaluating model performance. The model's effectiveness is evaluated after training, and the final accuracy and loss are given using testing data. Additionally, the trained model is stored in a h5 file for later usage. The hyperparameters like batch size and learning rate were finetuned using both grid search and then coarse to fine search.

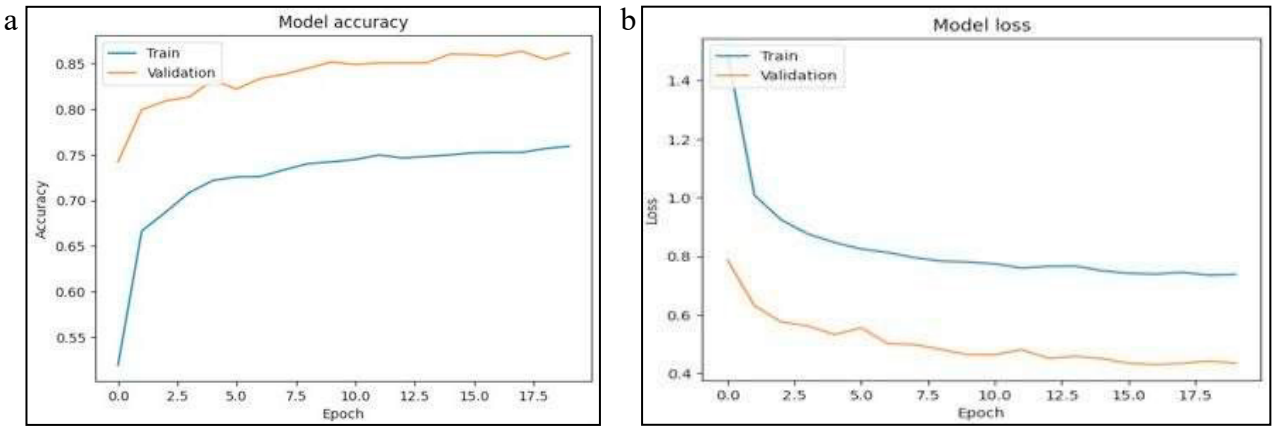


Fig. 4. Plot of (a) Model Accuracy (b) Model Loss with Epochs.

The model is transformed into a format that can be used on mobile devices to deploy the trained model to produce a mobile app. One typical method is to transform the model into the TensorFlow Lite format, which was created

especially for embedded and mobile devices. On mobile devices, TensorFlow Lite offers effective inference and supports a number of optimisations that both shrink the model's size and boost speed.
Using the cross-platform mobile app development framework Flutter, we will integrate the model into the mobile app after converting it to TensorFlow Lite format. With the help of these frameworks, you can load and execute the TensorFlow Lite model inside of a mobile application and use it to forecast the contents of pictures the app has taken or that you've chosen from the device's photo library. The app can then show the outcomes of the predictions made by the algorithm, such as classifying the type of plant disease from a plant photograph. The model would be integrated into a mobile app using the Flutter mobile app development technology and delivered to a mobile app by converting it to TensorFlow Lite format.

5. Results and Discussion

The suggested convolution neural network achieved an overall testing accuracy of 86.21% using data augmentation and transfer learning. The improvement in accuracy and loss over time from epoch 13 to 20 is seen in Table III. The accuracy improves, and the loss is significantly reduced with each epoch. The results have been reported for 20 epochs due to system limitations. Although the loss value continues to drop significantly after iteration 10, there has been no appreciable gain in our model's accuracy.

Table 3. Variation in Accuracy and Loss value with Epochs

Epoch Number	Accuracy (%)	Loss Value
13	85.09	4.515
14	85.09	4.579
15	86.06	4.504
16	86.02	4.343
17	85.86	4.295
18	86.38	4.340
19	85.48	4.420
20	86.21	4.344

Furthermore, table 3 demonstrates that as more iterations are performed, the value of loss and accuracy gets closer to being constant with different epochs.

Over 20 training epochs, the maximum validation accuracy of 74.94% was recorded, while the best training accuracy of 86.21% was also noted. A validation accuracy of 86% on average has been found. This is a helpful measure of how accurately the Deep Learning algorithm has classified various objects. A method of visualising and indicating the rate of model convergence is provided by the plots of train and test accuracy and loss against the epochs in Fig. 4. It is clear that the model has stabilised after about 20 epochs, and the metrics do not indicate that the model has much improved in the last ten epochs. In Figure 5, The images are given as input to the trained model, and the predicted labels match with the actual label. The outcomes demonstrate the model's effectiveness on the dataset and its applicability for classifying leaf diseases.

The proposed CNN model architecture leverages the hierarchical and translation-invariant nature of CNNs to extract meaningful features from plant disease images. The model can effectively detect and classify plant diseases by combining convolutional, pooling, and fully connected layers. However, further experimentation and optimization may be necessary to finetune the architecture for specific datasets and disease detection tasks.

Fig. 5. Prediction of diseases using CNN model. The label category denotes the predicted disease of the respective plant

The performance of the Convolutional Neural Network (CNN) and K-Nearest Neighbors (KNN) on the Plant Village dataset was evaluated. The accuracy achieved by both models was initially evaluated. The proposed Convolutional Neural Network (CNN) achieved an accuracy of 86.21%, and K-Nearest Neighbors (KNN) achieved an accuracy of 82.5%. The CNN's higher performance appears to be significantly influenced by its capacity to capture complex characteristics and patterns in the images of plant disease. The KNN classifier, in comparison, may perform poorly due to its dependency on the selection of K and sensitivity to the distribution of the dataset.
Its applicability for challenging picture classification tasks, such as plant disease identification, is further supported by the CNN's generalisation capabilities, which enables it to adapt to new and unexplored data. In conclusion, our comparison analysis unequivocally shows that, in terms of accuracy and overall performance, the CNN-based technique is superior to the KNN classifier for plant disease identification. This emphasises how crucial it is to use deep learning methods, such as CNNs, for reliable and precise disease identification in agriculture.

In summary, Convolutional layers, pooling layers, and fully connected layers are used to extract and classify disease-related information from input images in the proposed CNN model architecture, which, in turn, offers a framework for precise disease detection in plants.

6. Conclusion and Future Scope

The majority of the Indian people continue to rely heavily on the agricultural industry, which is still one of the most significant sectors. Therefore, spotting diseases in these crops is essential to expanding the economy. Therefore, this research aims to discover and characterize distinct illnesses in the tomato, potato and pepper bell crops. Using a Convolutional Neural Network model, the suggested method classifies tomato leaf diseases, potato leaf diseases, and pepper bell leaf diseases from the Plant Village dataset. To categorize tomato, potato and pepper bell leaf illnesses into different classes, a straightforward Convolutional Neural Network with a minimal number of layers was utilized as the architecture.

Future Scope: Future work may also involve experimenting with the suggested model using various learning rates and optimizers. It can also involve experiments with more modern architectures to improve the model's functionality on the train set. As a result, farmers can utilize the aforementioned model as a decision-making tool to assist and support them in recognizing diseases that can affect tomato, potato and pepper bell plants.

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